Dynamical Lie Algebra Framework for Quantifying Monotonic QAOA Schedules

A Theoretical Report

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1. Introduction and Motivation

The Quantum Approximate Optimization Algorithm (QAOA) applies a depth-p alternating circuit to an initial state $|+\rangle^{\otimes N}$, with $2p$ variational parameters $(\boldsymbol{\beta}, \boldsymbol{\gamma}) \in [0, 2\pi)^{2p}$. As p increases, the expressivity of the circuit grows, and one expects the achievable approximation ratio to improve.

A natural question arises: **if we draw all $2p$ parameters uniformly at random and evaluate the prefix expectation values $S_1, S_2, \ldots, S_p$ (using the first k layers for each S_k), how likely is it that this sequence is monotonically increasing?**

In other words, how rare is it for a random QAOA schedule to exhibit systematic improvement at every depth? Answering this question provides insight into:

- The **trainability** of QAOA: if monotonic schedules are exponentially rare, naive random initialization is unlikely to find them.
- The **structure of the QAOA landscape**: the decay rate of monotonic schedules encodes information about how quickly the circuit scrambles quantum states.
- The **role of the mixer**: different mixers (X-mixer, Grover mixer) generate different Lie algebras, leading to quantitatively different scrambling rates and monotonicity proportions.

We argue that the correct theoretical tool for analyzing this question is the **Dynamical Lie Algebra (DLA)** of the QAOA generators, combined with **concentration of measure** on the resulting reachable set of unitaries.

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2. Preliminaries: QAOA and Prefix Schedules

2.1 The QAOA Circuit

Given an N-qubit cost Hamiltonian H_C (diagonal in the computational basis) and a mixer Hamiltonian H_B, the QAOA unitary at depth p is:

$$U_p(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \prod_{k=1}^{p} e^{-i\beta_k H_B}\, e^{-i\gamma_k H_C}$$

The expectation value of H_C in the QAOA state is:

$$\langle H_C \rangle_p = \langle + |^{\otimes N}\, U_p^\dagger\, H_C\, U_p\, | + \rangle^{\otimes N}$$

2.2 Prefix Expectation Values

For a full depth-p parameter schedule $(\beta_1, \ldots, \beta_p, \gamma_1, \ldots, \gamma_p)$, define the **prefix expectation value** at depth $k \leq p$:

$$S_k \equiv S_k(\beta_{1:k}, \gamma_{1:k}) = \langle + | , U_k^\dagger(\beta_{1:k}, \gamma_{1:k}) , H_C , U_k(\beta_{1:k}, \gamma_{1:k}) , | + \rangle$$

where U_k uses only the first k layers. Crucially, each S_k depends on the same angles as S_{k-1} plus two additional parameters (β_k, γ_k).

2.3 The Score Convention

For minimization problems (e.g., SK model), we define $\text{score} = -\langle H_C \rangle$ so that "monotonically increasing score" corresponds to "getting closer to the ground state energy." For maximization problems (e.g., MaxCut), score $= \langle H_C \rangle$ directly. In either case, S_k denotes the score at depth k, and we seek $S_2 > S_1 > S_0$ etc.

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3. Formal Problem Statement

3.1 Monotonicity Survival Probability

Definition. The *monotonic-increase survival probability* at depth p is:

$$\boxed{f_{\uparrow}(p) \;=\; \Pr_{\beta,\gamma\sim\mathrm{Unif}[0,2\pi)^{2p}}\left[\,\forall\, k \in \{2,\ldots,p\}:\;\; S_k > S_{k-1}\,\right]}$$

Analogously, $f_{\downarrow}(p)$ is the probability that the sequence is strictly decreasing. Our experiment measures both, but the theoretical treatment focuses on $f_{\uparrow}(p)$ without loss of generality.

3.2 Chain-Rule Decomposition

Using the product rule of conditional probability:

$$f_{\uparrow}(p) \;=\; \prod_{k=2}^{p} q_k$$

where the **conditional per-step improvement probability** is:

$$q_k \;=\; \Pr[S_k > S_{k-1} \mid S_j > S_{j-1} \;\; \forall\, 2 \le j \le k-1]$$

Intuition: q_k asks "given that the schedule has been monotonically improving up through depth $k-1$, what is the probability that adding one more random layer still improves the score?" The overall survival probability $f_{\uparrow}(p)$ is the product of surviving each individual step.

3.3 The Central Question

What governs q_k, and how does it depend on:

- The depth index k?
- The number of qubits N?
- The choice of mixer H_B?
- The choice of cost Hamiltonian H_C?

The DLA framework provides a unified answer.

4. The Dynamical Lie Algebra (DLA)

4.1 Definition

Given a set of Hermitian generators $\{H_1, H_2, \ldots\}$, the **Dynamical Lie Algebra** is the real Lie algebra generated by the skew-Hermitian operators $\{iH_1, iH_2, \ldots\}$ under the matrix commutator bracket:

$$\mathfrak{g} = \langle iH_1, iH_2, \ldots \rangle_{\text{Lie}}$$

Concretely, $\mathfrak{g}$ is the smallest real vector space containing all iH_j that is closed under the commutator:

$$A, B \in \mathfrak{g} \implies [A, B] = AB - BA \in \mathfrak{g}$$

4.2 Construction by Iterated Commutators

Starting from the generators $\mathcal{S}_0 = \{iH_C, iH_B\}$:

$$\mathcal{S}_1 = \mathcal{S}_0 \cup \{[iH_C, iH_B]\}$$

$$\mathcal{S}_2 = \mathcal{S}_1 \cup \{[iH_C, [iH_C, iH_B]],\ [iH_B, [iH_C, iH_B]]\}$$

$$\vdots$$

Continue until $\text{span}(\mathcal{S}_{m+1}) = \text{span}(\mathcal{S}_m)$, at which point $\mathfrak{g} = \text{span}(\mathcal{S}_m)$.

4.3 The Reachable Set

The DLA determines the **reachable set** of unitaries:

$$\mathcal{R} = e^{\mathfrak{g}} = \{e^A \mid A \in \mathfrak{g}\}$$

This is a connected Lie group $G \subseteq \mathrm{SU}(2^N)$. Any QAOA circuit $U_p(\boldsymbol{\beta}, \boldsymbol{\gamma})$ lies in G, regardless of the parameter values.

Key fact: The dimension $d = \dim(\mathfrak{g})$ governs the effective number of independent directions the circuit can explore. When d is large, the circuit can access a high-dimensional manifold of unitaries; when d is small, it is constrained to a low-dimensional submanifold.

4.4 Intuitive Analogy

Think of the QAOA circuit as a random walk on the Lie group G. Each layer $(e^{-i\beta_k H_B} e^{-i\gamma_k H_C})$ takes a random step. The DLA dimension d is the dimensionality of the "room" in which the walk takes place:

- **Large d (e.g., $d = 4^N - 1$):** The room is enormous. After a few steps, the walker is lost in a featureless, high-dimensional space. Every direction looks the same, and the probability of continuing in any particular direction (like "improving the score") is essentially $1/2$.
- **Small d (e.g., $d = O(N^2)$):** The room is small. The walker remains near structured, low-dimensional regions for much longer. The landscape retains features, and the probability of improvement can stay appreciably above $1/2$.

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5. DLA for QAOA Generators

5.1 X-Mixer

The standard transverse-field mixer is:

$$H_B^{(X)} = \sum_{i=1}^{N} X_i$$

For a **generic** cost Hamiltonian H_C (one with no special symmetries, as is typical for random SK or MaxCut instances), the DLA generated by $\{iH_C, iH_B^{(X)}\}$ is generically the full Lie algebra:

$$\mathfrak{g}_X = \mathfrak{su}(2^N)$$

with dimension:

$$\dim(\mathfrak{g}_X) = 4^N - 1$$

Why? The operator $H_B^{(X)} = \sum_i X_i$ is a sum of single-qubit operators. The cost Hamiltonian H_C for SK or MaxCut involves two-body $Z_i Z_j$ terms. The commutator $[iH_C, iH_B^{(X)}]$ generates terms like $Y_i Z_j$ (single-qubit rotation on site i, conditioned on site j). Further commutators generate all multi-qubit Pauli strings, eventually spanning the full $\mathfrak{su}(2^N)$.

More precisely, write $H_C = \sum_{(i,j)} J_{ij} Z_i Z_j + \sum_i h_i Z_i$. Then:

$$[iZ_i Z_j,\ iX_i] = [iZ_i, iX_i] \otimes Z_j = -2Y_i Z_j$$

$$[iY_i Z_j,\ iX_j] = Y_i \otimes [iZ_j, iX_j] = -2Y_i Y_j$$

Continuing this process generates all Pauli strings of increasing weight, eventually spanning $\mathfrak{su}(2^N)$ for generic coupling graphs. (Non-generic cases, such as highly symmetric coupling structures, may yield a proper subalgebra.)

5.2 Grover Mixer

The Grover mixer is the projector onto the uniform superposition:

$$H_B^{(\text{Grov})} = |+\rangle\langle+| = \frac{1}{2^N} \sum_{x,y \in \{0,1\}^N} |x\rangle\langle y|$$

This operator acts within the **symmetric subspace** under permutations of computational basis states. For a cost Hamiltonian H_C that is diagonal in the computational basis, the Grover mixer preserves the decomposition of the Hilbert space into sectors of equal Hamming weight.

More precisely, define the subspaces $\mathcal{H}_w = \text{span}\{|x\rangle : |x| = w\}$ for $w = 0, 1, \ldots, N$, where $|x|$ is the Hamming weight. The Grover mixer connects all $\mathcal{H}_w$ sectors (since $|+\rangle$ has components in all sectors), but when combined with a diagonal H_C, the resulting DLA acts block-diagonally in a basis adapted to these sectors.

The key result is:

$$\dim(\mathfrak{g}_{\text{Grov}}) = O(N^2)$$

Specifically, if H_C has the form $H_C = \sum_{(i,j)} J_{ij} Z_i Z_j$, then the number of distinct values of $\langle x|H_C|x\rangle$ as x ranges over all 2^N bitstrings determines the effective dimension. For generic couplings, the Grover mixer and diagonal H_C generate a DLA whose dimension scales as $O(L^2)$ where L is the number of distinct energy levels, which is at most 2^N but is effectively much smaller when the cost function has symmetries.

However, in the subspace picture where $H_B^{(\text{Grov})}$ is rank-1, the algebra is much more constrained. Using the formula from Larocca et al.:

$$\dim(\mathfrak{g}_{\text{Grov}}) \leq (N+1)^2 - 1$$

since the effective Hilbert space dimension is $N + 1$ (the number of distinct Hamming weight sectors).

5.3 Summary Table

Mixer	$\dim(\mathfrak{g})$	**Scaling**	**Effective Hilbert space dim**
X-mixer (generic H_C)	$4^N - 1$	Exponential	2^N

Mixer	$\dim(\mathfrak{g})$	Scaling	Effective Hilbert space dim
Grover mixer	$\leq (N+1)^2 - 1$	Polynomial	$N+1$

This exponential gap in DLA dimension is the fundamental reason the two mixers produce dramatically different monotonicity statistics.

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6. Concentration of Measure and Scrambling

6.1 Haar Measure on the Reachable Set

The Lie group $G = e^{\mathfrak{g}}$ carries a natural invariant (Haar) measure μ_G. As the QAOA depth p increases with random parameters, the distribution of U_p over G approaches the Haar measure. The rate of convergence depends on whether the parameterized circuit forms an **approximate t-design** on G.

Definition. A distribution ν over G is an ϵ-approximate t-design if, for all polynomial functions f of degree $\leq t$ on G:

$$|\mathbb{E}_{U\sim\nu}[f(U)] - \mathbb{E}_{U\sim\mu_G}[f(U)]| \leq \epsilon \cdot \|f\|$$

6.2 Design Depth

The **design depth** $p^*(t)$ is the minimum QAOA depth at which the random circuit ensemble forms an approximate t-design on G. For the quantities we care about (second moments of expectation values), $t = 2$ suffices.

For random circuits on a group of dimension $d = \dim(\mathfrak{g})$:

$$p^*(2) \sim \frac{d}{(\text{local parameters per layer})}$$

Since each QAOA layer has 2 parameters, we estimate:

$$p_X^*(2) \sim \frac{4^N - 1}{2} \approx 2^{2N-1}$$

$$p_{\text{Grov}}^*(2) \sim \frac{(N+1)^2 - 1}{2} \approx \frac{N^2}{2}$$

Physical meaning: For the X-mixer, the circuit doesn't look Haar-random until exponentially many layers — far beyond any practical QAOA depth. For the Grover mixer, Haar-randomness on its (small) reachable set is achieved after only $O(N^2)$ layers.

6.3 Lévy's Lemma (Concentration of Measure)

Once the circuit approximates the Haar measure on G, the expectation value $S_p = \langle\psi|H_C|\psi\rangle$ concentrates around its mean:

$$\mu = \mathbb{E}_{U\sim\mu_G}\left[\langle +|U^\dagger H_C U|+\rangle\right]$$

Lévy's lemma on the compact Lie group G states:

$$\Pr[|S_p - \mu| > \epsilon] \leq 2\exp\left(-\frac{c \cdot d_{\text{eff}} \cdot \epsilon^2}{\|H_C\|_{\text{op}}^2}\right)$$

where c is a universal constant and d_{eff} is an effective dimension related to $\dim(\mathfrak{g})$.

Interpretation: In the scrambled regime, the expectation value is exponentially concentrated (for the X-mixer, since $d_{\text{eff}} \sim 4^N$) around $\mu = \mathrm{Tr}(H_C)/2^N$, which is the expectation value of the maximally mixed state. The landscape of S_p as a function of the parameters becomes essentially flat.

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7. Connecting DLA Dimension to Monotonicity Decay

7.1 The Per-Step Improvement in the Scrambled Regime

Consider step k in the QAOA schedule. The prefix state is:

$$|\psi_{k-1}\rangle = U_{k-1}(\beta_{1:k-1}, \gamma_{1:k-1}) |+\rangle$$

and the next-step expectation value is:

$$S_k(\beta_k, \gamma_k) = \langle \psi_{k-1} | \, e^{i\gamma_k H_C} e^{i\beta_k H_B} \, H_C \, e^{-i\beta_k H_B} e^{-i\gamma_k H_C} \, | \psi_{k-1} \rangle$$

This is a smooth function of the two random variables $(\beta_k, \gamma_k) \in [0, 2\pi)^2$.

In the scrambled regime (when k is large enough that $|\psi_{k-1}\rangle$ is approximately Haar-random on the reachable set), S_k and S_{k-1} are both drawn from approximately the same marginal distribution, concentrated around μ.

7.2 Variance of the Expectation Value Landscape

The key quantity is the variance of S_k as a function of (β_k, γ_k), averaged over the (scrambled) prefix:

$$\sigma_S^2 = \mathrm{Var}_{(\beta_k, \gamma_k),\, |\psi_{k-1}\rangle \sim \mu_G} [S_k]$$

From the barren plateau literature (Larocca et al., 2022; Ragone et al., 2024), this variance is controlled by the DLA dimension:

$$\sigma_S^2 \sim \frac{\|H_C\|_{\mathrm{HS}}^2}{\dim(\mathfrak{g})}$$

where $\|H_C\|_{\mathrm{HS}}^2 = \mathrm{Tr}(H_C^2)/2^N$ is the (normalized) Hilbert-Schmidt norm.

7.3 From Variance to Improvement Probability

Model S_{k-1} and S_k as jointly distributed random variables with:

- Marginal mean: μ
- Marginal variance: σ_S^2
- Correlation: ρ (arising from the shared prefix parameters)

Then the improvement probability is:

$$q_k = \Pr[S_k > S_{k-1}]$$

For the difference $\Delta_k = S_k - S_{k-1}$:

- $\mathbb{E}[\Delta_k] = 0$ (both have the same marginal mean)
- $\mathrm{Var}[\Delta_k] = 2\sigma_S^2(1-\rho)$

If Δ_k is approximately symmetric around zero (which follows from the approximate symmetry of the Haar distribution):

$$q_k \approx \frac{1}{2}$$

The corrections to $q_k = 1/2$ come from:

1. **Skewness** of the distribution of Δ_k, which is $O(1/\sqrt{d})$.
2. **Conditioning** on survival to step $k-1$ (the monotonicity constraint biases the distribution of $|\psi_{k-1}\rangle$).

To leading order:

$$\boxed{q_k = \frac{1}{2} + O\left(\frac{1}{\sqrt{\dim(\mathfrak{g})}}\right)}$$

7.4 The Exponential Decay Law

If $q_k \approx q$ is approximately constant for k in the scrambled regime, then:

$$f_\uparrow(p) = \prod_{k=2}^{p} q_k \approx q^{p-1}$$

This gives **exponential decay**:

$$\boxed{f_\uparrow(p) \approx A\, e^{-b\,p}, \qquad b = -\ln q, \qquad A = 1/q}$$

Since $q \leq 1/2 + O(d^{-1/2})$, the decay rate satisfies:

$$b \geq \ln 2 - O\left(\frac{1}{\sqrt{\dim(\mathfrak{g})}}\right)$$

For the X-mixer with $\dim(\mathfrak{g}) = 4^N - 1$:

$$b_X \approx \ln 2 - O(2^{-N}) \approx \ln 2 \approx 0.693$$

For the Grover mixer with $\dim(\mathfrak{g}) \leq (N+1)^2 - 1$:

$$b_{\text{Grov}} \approx \ln 2 - O(1/N)$$

The correction $O(1/N)$ for the Grover mixer is **polynomially larger** than the correction $O(2^{-N})$ for the X-mixer, so the Grover mixer should show noticeably slower decay, especially at small N.

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8. Regime Analysis

The derivation in Section 7 assumes the system is in the "scrambled regime." In practice, the QAOA circuit passes through three distinct regimes as depth k increases:

Regime I: Structured ($k \ll k^*$)

The state $|\psi_k\rangle$ is still close to $|+\rangle$ (the initial state). The expectation value landscape is highly structured, with large gradients and strong parameter dependence. The improvement probability q_k is typically high — the landscape has clear "downhill" directions.

Characterization: Perturbation theory applies. To first order in β_1 and γ_1:

$$S_1 \approx \mu + 2\beta_1 \sum_i \langle +|[H_B, H_C]|+\rangle + O(\beta_1^2, \gamma_1^2)$$

At depth $k = 1$, the expectation value is a low-degree trigonometric polynomial in (β_1, γ_1), with $O(1)$ variance.

Regime II: Transition ($k \sim k^*$)

The state begins to explore more of the reachable set. The landscape flattens, σ_S^2 decreases, and q_k drops from its initial high value toward $1/2$. In this regime, q_k depends on k, producing deviations from pure exponential decay.

If q_k decreases linearly with k in this regime, $\ln q_k \approx -\alpha k$, and:

$$\ln f_\uparrow(p) \approx -\sum_{k=2}^{p} \alpha k \approx -\frac{\alpha}{2} p^2$$

yielding **Gaussian/super-exponential** decay:

$$f_\uparrow(p) \sim A \exp(-b\, p^2)$$

Regime III: Scrambled ($k \gg k^*$)

The state is approximately Haar-random on G. The per-step improvement probability saturates at $q_\infty \approx 1/2$, and $f_\uparrow(p)$ decays as a pure exponential.

Crossover Depth k^*

The crossover from structured to scrambled occurs at:

$$k^* \sim \frac{\dim(\mathfrak{g})}{2}$$

since each layer contributes 2 free parameters to exploring the d-dimensional manifold.

Mixer	k^*	**Observable in experiments?**
X-mixer	$\sim 2^{2N-1}$	**No** (exponentially large; all experimental depths are in Regime I/II)
Grover mixer	$\sim N^2/2$	**Yes** (for $N = 4$: $k^* \approx 8$; for $N = 10$: $k^* \approx 50$)

Crucial implication: For the X-mixer at any practical QAOA depth ($p \leq 20$), the system is always in Regime I or II. The landscape is not yet flat, and the decay of $f_\uparrow(p)$ may be faster or slower than pure exponential, depending on how quickly the structured landscape transitions. For the Grover mixer, you may observe the full transition from structured to scrambled within your experimental range.

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9. Mixer-Specific Predictions

9.1 X-Mixer: Deep in the Structured Regime

Since $k^*_X \sim 2^{2N-1} \gg p$ for all practical depths:

- The landscape retains structure at all depths you probe.
- q_k may decrease with k as the state gradually explores more of $\mathrm{SU}(2^N)$.
- **Prediction:** $f_\uparrow(p)$ decays faster than pure exponential (super-exponential behavior). Best-fit model: exp_quadratic or stretched_exp.

9.2 Grover Mixer: Approaching Scrambled

Since $k^*_{\text{Grov}} \sim N^2/2$ is within reach:

- For small N (4–6 qubits), $k^* \approx 8$–18. Within your $p \leq 20$ range, you should see the transition to the scrambled regime.
- **Prediction:** $f_\uparrow(p)$ shows a transition from faster-than-exponential decay at small p to approximately exponential decay at larger p. Best-fit model: stretched_exp (which can interpolate between super-exponential and exponential behavior).

9.3 Quantitative Prediction: Grover Decays Slower

At any given p and N:

$$f_\uparrow^{(\text{Grov})}(p) \;>\; f_\uparrow^{(X)}(p)$$

The ratio should grow with N (since $\dim(\mathfrak{g}_X)$ grows exponentially while $\dim(\mathfrak{g}_{\text{Grov}})$ grows polynomially).

9.4 Scaling of the Decay Rate b with N

Define the effective decay rate from a fit $f_\uparrow(p) \approx Ae^{-bp}$:

$$b_X(N) \to \ln 2 \quad \text{as } N \to \infty \quad \text{(exponentially fast)}$$

$$b_{\text{Grov}}(N) \to \ln 2 \quad \text{as } N \to \infty \quad (\text{polynomially: } \ln 2 - c/N)$$

Measuring $b(N)$ for both mixers and plotting the deviation $\ln 2 - b(N)$ versus N would directly reveal the scaling, providing a quantitative test of the DLA theory.

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10. Problem-Hamiltonian Dependence

10.1 SK Model

The Sherrington-Kirkpatrick Hamiltonian is:

$$H_C^{(\text{SK})} = \sum_{i<j} J_{ij} Z_i Z_j, \qquad J_{ij} \sim \mathcal{N}(0, 1/N)$$

This is a fully connected Ising model with random couplings. The spectrum of $H_C^{(\text{SK})}$ has:

- Width: $\|H_C\|_{\text{HS}} = O(\sqrt{N})$
- Level spacing: exponentially small in N

For the X-mixer, the all-to-all connectivity of J_{ij} ensures that $\mathfrak{g} = \mathfrak{su}(2^N)$ with probability 1 (over the random couplings).

10.2 MaxCut

For an unweighted MaxCut problem on a random Erdős-Rényi graph $G(N, p_{\text{edge}})$:

$$H_C^{(\text{MaxCut})} = \frac{1}{2} \sum_{(i,j)\in E} (I - Z_i Z_j)$$

The structure is similar to SK but with $J_{ij} \in \{0, 1\}$ rather than Gaussian. For generic random graphs, the DLA is still $\mathfrak{su}(2^N)$ for the X-mixer.

10.3 Expected Differences

The main difference between SK and MaxCut lies in the **spectral statistics** of H_C:

- SK has a denser, more continuous spectrum with less degeneracy.
- MaxCut has a discrete spectrum with possible degeneracies (especially for small or structured graphs).

These affect the variance σ_S^2 and hence the fine details of q_k, but the DLA dimension (which controls the dominant scaling) is the same for both problems with the X-mixer.

Prediction: At fixed N and mixer, SK and MaxCut should show similar decay rates b, differing primarily in the prefactor A and in the variance across random instances.

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11. Comparison with i.i.d. Ordering Bound

11.1 The Reference Model

If $S_1, S_2, \ldots, S_p$ were independent and identically distributed continuous random variables, the probability of any specific strict ordering is:

$$f_{\text{iid}}(p) = \frac{1}{p!}$$

By Stirling's approximation:

$$\ln f_{\text{iid}}(p) \approx -p \ln p + p - \frac{1}{2} \ln(2\pi p)$$

11.2 Interpretation of the Ratio $f_\uparrow(p)/f_{\text{iid}}(p)$

Define the **correlation boost**:

$$R(p) = f_\uparrow(p) \cdot p!$$

- $R(p) > 1$**:** Parameter-sharing correlations make monotonicity **more likely** than the i.i.d. baseline.
- $R(p) < 1$**:** Correlations make monotonicity **less likely**.
- $R(p) = 1$**:** Consistent with effectively independent prefix evaluations.

Expected behavior:

- At small p, correlations are strong (consecutive S_k share most parameters), so $R(p)$ should be significantly greater than 1.
- As p grows, the incremental information in (β_k, γ_k) becomes a smaller fraction of the total, and correlations between consecutive S_k values increase, potentially keeping $R(p) > 1$.

However, the conditioning on monotonicity survival (which selects increasingly atypical parameter configurations) complicates this picture. Measuring $R(p)$ empirically is an informative diagnostic.

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12. Computable DLA Dimensions for Small Instances

For your small problem instances, you can directly compute $\dim(\mathfrak{g})$ and correlate it with the measured decay rate. Here is the algorithm:

12.1 Numerical DLA Computation

```
Input: Hermitian matrices H_C, H_B (dimension 2^N × 2^N)
Output: dim(g)
```

```
1. Initialize basis B = {iH_C, iH_B}
2. Orthonormalize B (e.g., Gram-Schmidt with Hilbert-Schmidt inner product)
3. Queue Q = all pairs (A, B) with A, B ∈ B
4. While Q is non-empty:
   a. Pop (A, B) from Q
   b. Compute C = [A, B]
   c. Project C onto span(B); compute residual R = C - proj_B(C)
   d. If ||R||_HS > tolerance:
      - Normalize R and add to B
      - Add all pairs (R, B_i) to Q for B_i ∈ B
5. Return |B|
```

12.2 Expected Results

N	$\dim(\mathfrak{su}(2^N))$	$\dim(\mathfrak{g}_X)$ **(expected)**	$\dim(\mathfrak{g}_{\text{Grov}})$ **(upper bound)**
4	255	255	24
5	1023	1023	35
6	4095	4095	48
8	65535	65535	80
10	1048575	1048575	120

12.3 The Key Plot

After running your experiments, create a scatter plot of:

$$(\text{measured decay rate } b) \quad \text{vs.} \quad \log_2(\dim(\mathfrak{g}))$$

with points labeled by (mixer, N). If the theory is correct, there should be a clear monotone relationship: larger DLA dimension → faster decay (larger b) → rarer monotonic schedules.

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13. Summary of Testable Predictions

#	Prediction	Source	Experimentally testable?
1	$f_\uparrow(p)$ decays exponentially or super-exponentially in p	Chain rule + DLA concentration	Yes (fit models)
2	Decay rate $b_{\text{Grov}} < b_X$	$\dim(\mathfrak{g}_{\text{Grov}}) \ll \dim(\mathfrak{g}_X)$	Yes (compare fits)
3	$b_X(N) \to \ln 2$ exponentially fast in N	Concentration on $\mathfrak{su}(2^N)$	Yes (measure b_X for $N = 4, 6, 8, 10$)
4	$b_{\text{Grov}}(N) \to \ln 2$ polynomially in N	Concentration on restricted DLA	Yes
5	Grover mixer shows transition from super-exp to exp at $p \sim N^2/2$	Design depth scaling	Yes (observe curvature change on log-y plot)
6	X-mixer shows no such transition (always in structured regime)	$k_X^* \gg p_{\max}$	Yes
7	SK and MaxCut have similar b at fixed N, mixer	Same DLA, different spectrum	Yes (compare across problems)
8	Per-step q_k is measurable and should approach $1/2$ from above	Survival analysis	Yes (Kaplan-Meier estimator)
9	$f_\uparrow(p) \cdot p! > 1$ (correlations help monotonicity)	Parameter sharing	Yes (compute $R(p)$)

#	Prediction	Source	Experimentally testable?
10	b correlates monotonically with $\log(\dim(\mathfrak{g}))$	DLA theory	Yes (compute DLA for small N)

14. References

1. **Larocca, M., Czarnik, P., et al.** "Diagnosing Barren Plateaus with Tools from Quantum Optimal Control." *Quantum* 6, 824 (2022). — Establishes $\dim(\mathfrak{g})$ as the controlling quantity for variational quantum circuit expressibility and gradient variance.
2. **Ragone, M., Bakalov, B.N., et al.** "A Unified Theory of Barren Plateaus for Deep Parametrized Quantum Circuits." *Nature Communications* 15, 7960 (2024). — Extends the DLA framework to a general unified theory with precise variance bounds.
3. **Brandão, F.G.S.L., Harrow, A.W., Horodecki, M.** "Local Random Quantum Circuits are Approximate Polynomial-Designs." *Communications in Mathematical Physics* 346, 397–434 (2016). — Proves that random circuits converge to approximate t-designs at a rate controlled by the circuit architecture.
4. **D'Alessandro, D.** *Introduction to Quantum Control and Dynamics*. Chapman & Hall/CRC, 2007. — Textbook treatment of Lie-algebraic methods in quantum control, including DLA construction and reachability.
5. **Farhi, E., Goldstone, J., Gutmann, S.** "A Quantum Approximate Optimization Algorithm." *arXiv:1411.4028* (2014). — Original QAOA proposal.
6. **Ledoux, M.** *The Concentration of Measure Phenomenon*. AMS Mathematical Surveys and Monographs, Vol. 89, 2001. — Comprehensive reference for Lévy's lemma and concentration inequalities on manifolds.
7. **Harrow, A.W., Low, R.A.** "Random Quantum Circuits are Approximate 2-designs." *Communications in Mathematical Physics* 291, 257–302 (2009). — Convergence rates for random

circuits to form unitary designs.

End of report.